

Numerical Analysis Exercise 1

3. Show that $f'(x)$ is 0 at least once in the given intervals

(a) $f(x) = 1 - e^x + (e - 1) \sin \frac{\pi}{2} x, [0, 1];$

Proof there are $f(0) = f(1) = 0$,

According to the **Rolle's Theorem**

then, a number c in $(0,1)$ exists with

$$f'(c) = 0.$$

So that $f'(x)$ is 0 at least once in the given intervals.

(b) $f(x) = x \sin \pi x - (x - 2) \ln x, [1, 2];$

Proof: there are $f(1) = f(2) = 0$,

According to the **Rolle's Theorem**

then, a number c in $(1,2)$ exists with

$$f'(c) = 0.$$

So that $f'(x)$ is 0 at least once in the given intervals.

(c) $f(x) = (x - 2)(\sin x) \ln(x + 2), [-1, 3];$

Proof: there are $f(-2) = f(2) = 0$,

According to the **Rolle's Theorem**

then, a number c in $(-2,2)$ exists with

$$f'(c) = 0$$

So that $f'(x)$ is 0 at least once in the given intervals.

4. Find the fourth Taylor polynomial $P_4(x)$ for the function $f(x) = xe^x$ about $x_0 = 0$.

(a) Find an upper bound for $|f(x) - P_4(x)|$, for $0 \leq x \leq 0.4$.

Solution: According to the Taylor theorem

Thus we get

$$P_4(x) = f(0) + f'(0)x + \frac{f''(0)}{2!}x^2 + \frac{f'''(0)}{3!}x^3 + \frac{f^{(4)}(0)}{4!}x^4$$

we easily calculate

$$f(0) = 0, f'(0) = 1, f''(0) = 2, f'''(0) = 3, f^{(4)}(0) = 4$$

So we have

$$P_4(x) = x + x^2 + \frac{1}{2}x^3 + \frac{1}{6}x^4.$$

Then we assume

$$h(x) = |f(x) - P_4(x)| = xe^x - x - x^2 - \frac{1}{2}x^3 - \frac{1}{6}x^4$$

After we calculate by using **MatLab**, we get the max of h(x) is **4.6321 e-04**

```
1 x=0:0.01:0.4;
2 y=x.*exp(x)-x-x.^2-1/2*x.^3-1/6*x.^4;
3 max(y)
```

(b) Approximate $\int_0^{0.4} f(x)dx$ **using** $\int_0^{0.4} P_4(x)dx$.

Solution: by using the result of (a)

We get

$$P_4(x) = x + x^2 + \frac{1}{2}x^3 + \frac{1}{6}x^4.$$

Thus,

$$\int_0^{0.4} f(x)dx \approx \int_0^{0.4} P_4(x)dx = \int_0^{0.4} (x + x^2 + \frac{1}{2}x^3 + \frac{1}{6}x^4)dx = 0.104874666666667$$

calculate code as follows

```
1 syms x
2 ans=int(x+x^2+1/2*x^3+1/6*x^4,0,0.4);
3 vpa(ans,15)
```

6. Perform the following computation: (i) exactly,(ii) using three-digit chopping arithmetic, and (iii) using three-digit rounding arithmetic. Compute the relative errors in parts(ii) and (iii).

(a) $\frac{4}{5} + \frac{1}{3}$

(i) $\frac{17}{15}$ (ii) 1.13 (iii) 1.13

The relative errors in part (ii) is **0.2024**

The relative errors in part (iii) is **0.2024**

(b) $\frac{4}{5} \times \frac{1}{3}$

(i) $\frac{4}{15}$ (ii) 0.266 (iii) 0.266

The relative errors in part (ii) is **0.0025**

The relative errors in part (iii) is **0.0025**

(c) $(\frac{1}{3} - \frac{3}{11}) + \frac{3}{20}$

(i) $\frac{139}{660}$ (ii) 0.211 (iii) 0.210

The relative errors in part (ii) is **0.0019**

The relative errors in part (iii) is **0.0029**

(d) $(\frac{1}{3} + \frac{3}{11}) - \frac{3}{20}$

(i) $\frac{301}{660}$ (ii) 0.455 (iii) 0.456

The relative errors in part (ii) is **0.0023**

The relative errors in part (iii) is **1.3289 e-04**

7. Let $f(x) = \frac{x \cos x - \sin x}{x - \sin x}$

(a) Find $\lim_{x \rightarrow 0} f(x)$.

Solution By using the L'Hospital's rule

$$\begin{aligned}\lim_{x \rightarrow 0} f(x) &= \lim_{x \rightarrow 0} \frac{\cos x - x \sin x - \cos x}{1 - \cos x} = \lim_{x \rightarrow 0} \frac{-x \sin x}{1 - \cos x} = \lim_{x \rightarrow 0} \frac{-\sin x - x \cos x}{\sin x} = \lim_{x \rightarrow 0} \left(-1 - \frac{x \cos x}{\sin x}\right) \\ &= -1 - \lim_{x \rightarrow 0} \frac{\cos x}{\frac{\sin x}{x}} = -2\end{aligned}$$

Solution

(b) Use four-digit rounding arithmetic to evaluate $f(0.1)$.

Solution using four-digit rounding arithmetic, we have

$$\begin{aligned}\sin(0.1) &= 0.09983, \cos(0.1) = 0.9950 \\ f(0.1) &= \frac{0.1000 * 0.9950 - 0.09983}{0.1 - 0.09983} = -2.253\end{aligned}$$

(c) Replace each trigonometric function with its third Maclaurin polynomial and repeat part (b).

Solution

The third Maclaurin polynomial for $\sin x$ is as follows

$$\sin x = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!}$$

The third Maclaurin polynomial for $\cos x$ is as follows

$$\cos x = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \frac{x^6}{6!}$$

The Maclaurin polynomial for $f(x)$ is as follows,

$$\begin{aligned}f(x) &= \frac{x \cos x - \sin x}{x - \sin x} = \frac{x(1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \frac{x^6}{6!}) - (x - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!})}{x - (x - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!})} = \frac{-\frac{2x^3}{3} + \frac{x^5}{30} - \frac{8x^7}{5040}}{\frac{x^3}{3!} - \frac{x^5}{5!} + \frac{x^7}{7!}} \\ f(0.1) &= -1.500\end{aligned}$$

(d) The actual value is $f(0.1) = -1.99899998$. **Find the relative error for the values obtained in parts (b) and (c).**

the relative error for the values obtained in part (b)

$$e_r = \frac{|-1.99899998 + 2.253|}{|-1.99899998|} = 0.1271$$

the relative error for the values obtained in part (c)

$$e_r = \frac{|-1.99899998 + 1.500|}{|-1.99899998|} = 0.2496$$

8. Assume that the permissible relative error of the volume of a sphere is at most 10^{-2} . Determine the permissible relative error of its radius

Solution According to the formula for the volume of the ball,

$$V = \frac{4\pi r^3}{3}, \quad V^* = \frac{4\pi (r^*)^3}{3}$$

Since the relative error of the volume of a sphere is at most 10^{-2} .

$$e_r = \frac{|V - V^*|}{|V|} \leq 10^{-2}$$

So we have

$$|V - V^*| \leq 10^{-2}|V| \Rightarrow \frac{99}{100}V \leq V^* \leq \frac{101}{100}V \Rightarrow \frac{99}{100}r^3 \leq (r^*)^3 \leq \frac{101}{100}r^3$$

Thus,

$$\sqrt[3]{\frac{99}{100}}r \leq r^* \leq \sqrt[3]{\frac{101}{100}}r$$

9. The following approximations are all obtained by rounding arithmetic. Determine how many significant digits they have, respectively.

(a) $p_1^* = 1.1021$;

(b) $p_2^* = 0.031$;

(c) $p_3^* = 56.430$;

(d) $p_4^* = 7 \times 1.0$.

Solution Rewriting $p_1^*, p_2^*, p_3^*, p_4^*$ in the floating-point decimal form, respectively, gives

$$p_1^* = 0.11021 \times 10^1$$

$$p_2^* = 0.31 \times 10^{-1}$$

$$p_3^* = 0.56430 \times 10^2$$

$$p_4^* = 0.70 \times 10^1$$

Assume $p_1 = 1.10214$ or $p_1 = 1.102135$

$$|p_1 - p^*| = 0.00004 = 0.4 \times 10^{-4} \leq 5 \times 10^{-5}$$

$$|p_1 - p^*| = 0.000035 = 0.35 \times 10^{-4} \leq 5 \times 10^{-5}$$

the significant digits of p_1 is **5**.

In the same way

the significant digits of p_2 is **2**.

the significant digits of p_3 is **5**.

the significant digits of p_4 is **2**.